

Per-seed no-failure certificates for every value-based safe policy

And the two falsification attacks when a certificate fails

SPARK goal-insertion notes

What this note proves

For a fixed scene (seed) and a chosen safe policy, we want a certificate: a checkable condition under which the policy cannot fail anywhere the robot can reach. This generalizes the two we already did — SSA in `cx_derivation.pdf`, SSS in `sss_certificate.pdf` — to all of SPARK's value-based controllers, and then frames the two ways an attacker falsifies a failed certificate.

Throughout we reuse, from `cx_derivation`:

- $c_i(x) = \sum_k u_{lim,k} |(L_g \phi_i)_k|$ — the control-authority ceiling (fastest the motors can drive ϕ_i down), with reachable rate $\dot{\phi}_i \in [-c_i, +c_i]$;
- first-order dynamics, $f \equiv 0 \Rightarrow L_f \phi_i = 0$.

And from `cx_policies`, every value-based controller enforces the same row $\dot{\phi}_i \leq -d_i$ at active constraints, differing only in the demand d_i :

policy	demand d_i	active set	failure symptom
SSA	η	$\phi_i \geq 0$	hard QP infeasible $\rightarrow u_{ref}$
SSS	$\lambda \phi_i$	$\phi_i \geq 0$	hard QP infeasible $\rightarrow u_{ref}$
CBF	$\lambda \phi_i$	all masked	hard QP infeasible $\rightarrow u_{ref}$
r-SSA	η	$\phi_i \geq 0$	forced slack $s > 0$
r-SSS, r-CBF	$\lambda \phi_i$	as above	forced slack $s > 0$
p-SSA	η	$\phi_i \geq 0$	projection $s^* > 0$
SMA, PFM	—	$\phi_i \geq 0$	heuristic underbrake

The single quantity behind every certificate

Define the margin at a state, over the active constraints,

$$g(x) := \min_{i \text{ active}} (c_i(x) - d_i) \quad (L_f \phi = 0).$$

From `cx_derivation` Step 4 (and its multi-constraint Step 9), the policy's safety row is satisfiable at x iff $g(x) \geq 0$. So:

- Hard filters (SSA/SSS/CBF): $g(x) < 0 \iff$ QP infeasible $\iff$ give up $\rightarrow u_{ref}$.
- Soft filters (r-SSA/r-CBF/r-SSS): always solvable, but the slack is forced to be $\geq |g(x)|_+$ — a guaranteed violation of $\dot{\phi} \leq -d$.
- p-SSA: its min-norm projection equals exactly $|g(x)|_+$.

So for every value-based policy, “failure” (lost guarantee) at x is precisely $g(x) < 0$. This gives the universal direct certificate:

$$\boxed{\text{(direct)} \quad g(x) \geq 0 \text{ for all reachable active } x \implies \text{no failure (this policy, this seed).}}$$

The only thing that changes between policies is the demand d_i inside g , which makes the certificate easy or hard to discharge. The two demand models split here.

Family 1 — constant demand: SSA, r-SSA, p-SSA ($d = \eta$)

Here $g(x) = \min_i (c_i(x) - \eta)$, so the certificate is

$$\boxed{c_i(x) \geq \eta \text{ for all reachable active } x \implies \text{no failure.}}$$

The bar η is fixed and independent of the time step Δt . There is no boundary self-protection: at $\phi_i = 0^+$ the demand is still the full η , so a single reachable contact with $c_i(x) < \eta$ refutes the certificate. The irreducible vulnerable set is the shell $\{c(x) < \eta\}$ — it has positive thickness and does not shrink with Δt .

- SSA (hard): refutation $\implies$ a give-up $\rightarrow u_{ref}$ at that state.
- r-SSA (soft): refutation $\implies$ forced slack $\geq \eta - c$; with the large weight ($\rho = 10^3$) this manifests as a stall/deadlock, with small ρ as push-through.
- p-SSA (soft): refutation $\implies$ projection $s^* = (\eta - c)_+ > 0$; the certificate $g \geq 0$ is exactly “ $s^* \equiv 0$ everywhere reachable.”

For all three the discharge is identical: show every reachable active contact has $c(x) \geq \eta$. No Δt relief is available.

Family 2 — proportional demand: SSS, CBF, r-SSS, r-CBF ($d = \lambda\phi$)

Now $g(x) = \min_i (c_i(x) - \lambda\phi_i(x))$, certificate

$$\boxed{c_i(x) \geq \lambda\phi_i(x) \text{ for all reachable active } x \implies \text{no failure.}}$$

But the proportional demand vanishes at the boundary ($\phi = 0 \implies d = 0 \leq c$), which buys a forward-invariance upgrade (full derivation in `sss_certificate.pdf`):

1. one-step overshoot is bounded, $\dot{\phi}_i$ rises at most $c_i \Delta t$ per step (since $\dot{\phi}_i \leq c_i$);
2. the feasible recovery shell has thickness $\phi_i \leq c_i/\lambda$;
3. while active and feasible, SSS/CBF drive ϕ_i down ($\dot{\phi}_i \leq -\lambda\phi_i$, $\lambda\Delta t < 1$).

Chaining these, ϕ_i stays $\leq c_{\max}\Delta t$, and the certificate reduces to a single scalar:

$\text{start safe, } \lambda\Delta t < 1, \lambda\Delta t c_{\max} \leq c_{\min} \implies \text{no failure,}$	$c_{\min} = \inf_{\text{reach.}} c, c_{\max} = \sup_{\text{reach.}} c.$
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The bar now scales with Δt , so the irreducible vulnerable set collapses to $\{c(x) = 0\}$ — the controllability singularities of Step 8 — as $\Delta t \rightarrow 0$. Any seed whose reachable boundary keeps $c(x) > 0$ is certifiable for small enough Δt .

- SSS (hard, active only at $\phi \geq 0$): there is an inactive region $\phi < 0$ where u_{ref} runs free; the overshoot in (1) is what a single free step adds.
- CBF (hard, active everywhere): no inactive region — it regulates even at $\phi < 0$ (allowing $\dot{\phi} \leq -\lambda\phi = \lambda|\phi|$, i.e. a controlled approach), so in the continuous limit ϕ never reaches 0 from a safe start. Its failures are pure discretization/linearization overshoot, bounded by the same $c_{\max}\Delta t$.
- r-SSS, r-CBF (soft): same g , refutation $\implies$ forced slack $\geq |g|$.

Family 3 — heuristics: SMA, PFM (no sufficient certificate)

These solve no QP, so $g \geq 0$ is necessary for safety but not sufficient:

- SMA takes a fixed step $u = u_{ref} - c_{sma}L_g\phi_{\max}$ then clips to the box. Any box control obeys $\dot{\phi} \geq L_f\phi - c$, so $g < 0$ still forces failure (necessary), but the fixed step underbrakes relative to the optimal corner, so SMA also fails in part of $\{g \geq 0\}$. We can therefore refute (find failing states) but cannot certify with g alone.
- PFM maps a Cartesian push back through a pseudo-inverse without box clipping, so it may command unrealizable controls; $c(x)$ bounds the physics but not PFM's output. No clean certificate.

For these two we report the necessary-condition screen only.

Summary of certificates

family	policies	certificate to discharge	shrinks with Δt ?	irreducible set
constant η	SSA, r-SSA, p-SSA	$c(x) \geq \eta$ on reachable active set	no	$\{c < \eta\}$
proportional $\lambda\phi$	SSS, CBF, r-SSS, r-CBF	$c(x) \geq \lambda\phi(x)$, or $\lambda\Delta t c_{\max} \leq c_{\min}$	yes	$\{c = 0\}$

family	policies	certificate to discharge	shrinks with Δt ?	irreducible set
heuristic	SMA, PFM	none sufficient ($g \geq 0$ only necessary)	—	—

When the certificate fails: two falsification attacks

If the screen exhibits a reachable active state with $g(x) < 0$, the policy is not certifiable for that seed, and we realize the failure by steering the closed loop into such a state with an admissible inserted goal. Two distinct state-inducements, sharing the same objective ($\min_t g(x_t) < 0$) but different threat semantics:

Attack A — midpoint insertion G'_1 . Schedule $[G'_1, G_1]$: the robot detours through an extra waypoint G'_1 before the legitimate goal G_1 . Failure may occur anywhere along the detour, transiently. We fuzz admissible G'_1 to minimize $\min_t g(x_t)$ over the whole two-hop rollout, requiring G'_1 itself reachable. This is the goal-insertion attack of the earlier study — an extra spurious waypoint.

Attack B — initial-goal poisoning G'_0 . Schedule $[G'_0, G_1]$, but the objective measures $\min_t g(x_t)$ only on the second leg (after G'_0 is reached, `wp_idx` advanced to G_1). The robot first settles at an adversarial start G'_0 , then performs its normal task $G'_0 \rightarrow G_1$ — and that legitimate leg becomes infeasible. No extra waypoint is injected into the executed task; only the starting configuration is shifted. This is stealthier: the plan $G'_0 \rightarrow G_1$ looks ordinary, yet the safe controller fails on it.

Both are constrained to admissible goals (inside the workspace, outside obstacle keep-outs), so each leg is individually a legitimate reach; only the induced state is adversarial. A seed/policy is then reported as one of:

- CERTIFIED — screen found $g(x) \geq 0$ on all probed reachable states (empirical; see caveat);
- REFUTED, midpoint-realizable — Attack A drove $g < 0$;
- REFUTED, initial-realizable — Attack B drove $g < 0$ on the legitimate leg;
- REFUTED, not-realized — a low- g state exists but neither attack reached it within budget.

Caveat (same as SSA/SSS). “Reachable” is approximated by sampling rollouts (baseline + obstacle-biased one-hop probes), so CERTIFIED is an empirical certificate, not a formal proof; a formal proof needs a sound reachable-set over-approximation. The forward-invariance scalar $\lambda \Delta t c_{\max} \leq c_{\min}$ is reported alongside for the proportional family as the stronger (still sampling-bounded) witness.